Cycle-Consistent Audio-Visual Grouping

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Figure 1: **Cycle-consistent grouping.** Our model learns to associate image patches with sound using a grouping method based on cycle-consistent random walks. Here, we show sound source segmentation probabilities from our models, which have been trained on human speech and musical instruments datasets, respectively.

Abstract

In this paper, we address the problem of learning to group image patches into sound sources, using supervision provided only from raw audio-visual data. For this, we propose a method based on contrastive random walks [21]. We train a random walker to move along a graph whose nodes correspond to image patches and audio examples. The model is constrained to learn a cycle-consistency walk: associations that it learns should allow it to return to an image patch after transitioning across modalities to audio. The probability of the walker visiting a pair of examples on this walk then indicates the degree of similarity between them. We evaluate our method on human speech and musical instruments datasets. Experimental results show that, in comparison to other self-supervised and weakly supervised localization approaches, our model obtains significantly more complete segmentations, and learns a more effective measure of visual similarity.

1. Introduction

Today's methods for organizing scenes largely rely on labeled images provided by humans. By contrast, humans learn to do this organization without explicit supervision, and also do so using all of the sensory modalities available to them [32].

Recently, there have been a variety of methods that learn to group visual information via multi-modal self-supervision, such as identifying sound sources and detecting audio-visual objects [33, 4, 27, 2, 3]. These methods, however, are largely focused on the problem of *localization*: i.e. detecting *some*, but not all, of the pixels in the image that belong to the sound-making object. For example, existing sound source localization models, when applied to human speech, focus on detecting faces and mouths [1], without modeling other parts of the body. We desire more general grouping mechanisms that can convey *associations*. Such a model should be able to tell which two regions "go together" as part of the same object, and which image regions "go with" audio as part of a sound source.

In this paper, we learn similarity functions for sound source segmentation through self-supervision. We learn our similarity metric using a random walk on a graph, following the formulation of Jabri *et al*. [21]. In this graph, there are nodes for image patches and for audio, and edges that connect them. We train a random walker to walk from image patches to audio and then back. Transition probabilities for the walk are determined by computing dot products between learned embedding vectors. We learn embeddings that encourage the walk to traverse visual and audio examples that correspond, *i.e*. that were sampled from the same video, while also being cycle-consistent: it should maximize the probability of returning to the image patch that it originated from. After training, we measure the similarity between two image patches (and between an image patch and audio) by the probability that a random walk traverses them both.

This formulation has two advantages over other self-supervised localization methods. First, it explicitly learns a similarity function between image patches, thereby providing a grouping mechanism that operates on image patches alone, yet that is trained with multi-modal supervision. Sec-

ond, the loss encourages "many to one" matches between the image patches and audio, in contrast to multiple instance learning methods that potentially allow the model to ignore all but the most discriminative parts of the image.

We test our model on videos containing human speech [8] and of musical instruments [5]. We show that our method can learn better similarities between different visual regions compared to other methods. Besides, for sound source localization, we obtain visual segmentation that are significantly more complete than those of other self-supervised localization methods, and than those of weakly supervised object localization methods trained with class labels.

2. Related Work

Visual grouping. Methods aim to solve visual grouping are mainly unsupervised and focus only on visual data [20, 13, 24]. Greff *et al*. [13] propose a framework that combines neural networks and the EM algorithm to group and learn individual entities. Isola *et al*. [20] train a network to learn patch similarity by predicting whether two patches are the neighbors of each other. Haller *et al*. [14] use videos to do foreground/background segmentation, in this way utilizing "common fate" principle for grouping. Pathak *et al*. [29] use pseudo ground truth produced by unsupervised motion-based segmentation and train a network to segment objects. Different from the above methods, our method uses multi-modality data to perform grouping. We hope this can provides insight into exploring more cues for visual grouping.

Audio-visual learning. The recent surge of research in audio-visual learning focuses on the correspondence between video and audio. This includes directions: sound source localization [33, 4, 15, 28, 37, 2, 30, 19], sound source separation [1, 10, 12], audio-visual multi-sensory features [11, 27, 41] and audio-visual event parsing [36]. Our task is closely related to those trying to localize sound sources. These methods aim to localize the sound source in images by unitizing the correspondence between audio and image (or video) to perform self-supervised learning [4, 33, 3]. Generally, the proxy task is to predict whether an image-audio pair comes from an actual video, or a synthetically mismatched video. Arandjelovic and Zisserman [4, 3] measure the similarity between learned image region and audio representations, and use multiple instance learning to localize sound sources. Owens *et al*. use class activated map (CAM) [46] to visualize the area contributing to solving temporal shifting tasks. Inspired by [45], Rouditchenko *et al*. propose to segment objects using audio-visual correspondence [31] with category labels of videos. In contrast to this work, our goal is to obtain complete segmentations of sound sources, rather than approximate localization, and we don't need category labels. Moreover, in contrast to [45, 2], our goal is to learn from a single frame, rather than video, using object appearance alone.

Weakly supervised object localization. There are some works in weakly supervised object localization that share a similar objective as ours: discovering less discriminative parts of the object [35, 42, 43, 44, 7]. Although these methods require no pixel-wise ground-truth annotation of objects, they do require manual annotations such as bounding boxes, scribbles, or image-level labels. Most of these methods are trained on the base of the classification task and then rely on visualization methods, such as CAM, for localizing objects. Kolesnikov *et al*. [23] use a classification network to generate supervision signals and propose boundary loss to constrain the boundary of objects. Zhang *et al*. [43] propose to exploit less discriminative parts of the objects by adversarially erasing object regions. However, a more recent work [6] argues that the weakly supervised object localization task itself is ill-posed with only image-level labels. Inspired by this work, we use other forms of supervision—namely, the rich audio signal accompanying a video—to solve a problem akin to weakly supervised localization.

Graph representation learning. There are a number of works using graph to learn image/video segmentation [34] and visual correspondence [21], *etc*. Jabri *et al*. [21] propose to use graphs to learn the visual correspondence between several frames clipped from video, and the graph is constructed by connecting patches in spatio-temporal neighborhoods. While we adopt the random walk formulation from this work, we apply it to a multi-modal learning domain, rather than to video representation learning.

3. Method

Given paired images and audio, our goal is to learn similarities between patches for visual grouping. For a given example (an image with audio) we have image patches $\mathbf{x}_1, \mathbf{x}_2, ...\mathbf{x}_N$ and its corresponding sound $\mathbf{s}$. We'd like to learn a similarity function $\phi_I(\mathbf{x}_i, \mathbf{x}_j)$ that says how likely two image patches belong to the same sound source, as well as a similarity function $\phi_S(\mathbf{s}_i, \mathbf{x}_j)$ that says how likely an image patch belongs to the sound source that generated the audio $\mathbf{s}_i$.

3.1. Cross-modal random walks

We formulate these similarity functions as random walks on graphs. As a simple example, consider the problem of learning a similarity function between image patches by cross-modal prediction. One straightforward idea, closely related to self-supervised approaches [4, 3, 33], would be to predict audio for each image patches. The representation learned from this process — namely, the distribution of audio we'd expect to hear when seeing the patch — could then be used to measure similarities between image patches.

To learn our model, we create an image encoder that maps patches to feature vectors $f(\mathbf{x}_i)$, and an audio encoder g that

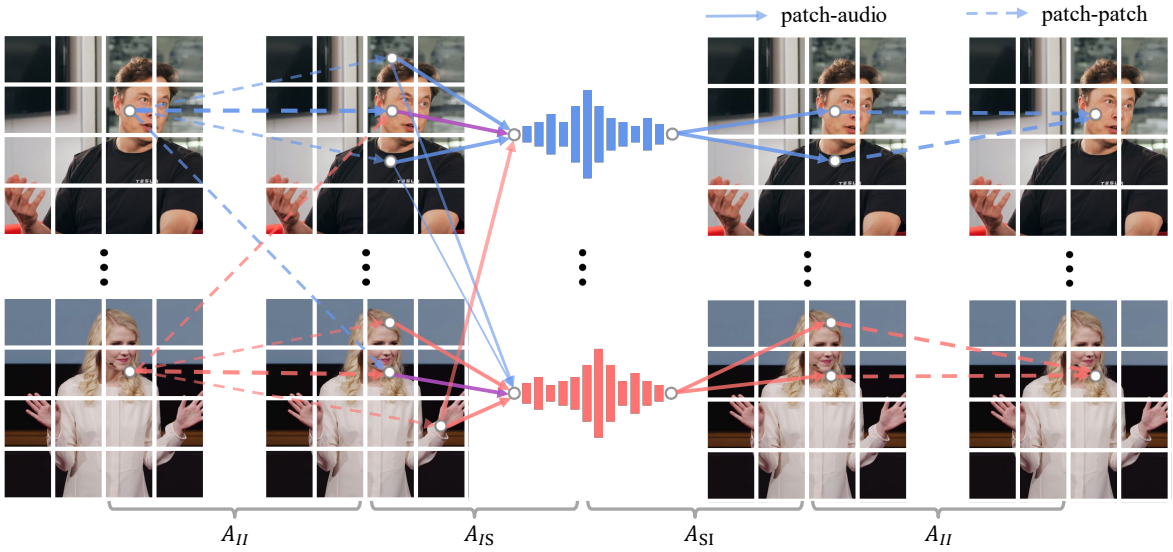


Figure 2: The overview of the proposed audio-visual random walk. A_{II} measures similarities between patches and patches while A_{IS} and A_{SI} measure similarities between audio and patches. A walk starts from a patch in the left. It travels to another image patch node, then visits an audio node following directed edges and finally reach a patch in the most right.

extracts audio feature vector $g(\mathbf{s})$. Following other works, we can learn this representation using a contrastive similarity function. Given $K-1$ other random *negative* audio, $f(\mathbf{x}_i)$ should have a high dot product with its corresponding audio $g(\mathbf{s})$, and low dot product with every other audio example. If $\tilde{A}_{IS}(i,j)$ is the similarity between i-th patch and j-th audio, these similarities can be formulated as:

$$\tilde{A}_{IS}(i,j) = \frac{\exp(f(\mathbf{x}_i)^\top g(\mathbf{s}_j)/\tau)}{\sum_{k=1}^{K}\exp(f(\mathbf{x}_i)^\top g(\mathbf{s}_k)/\tau)}, \tag{1}$$

where τ is a temperature hyper-parameter [40], and $\tilde{A}_{IS} \in \mathbb{R}^{N\times K}$. Following other works in contrastive learning, we can learn A_{IS} using a cross-entropy loss that encourages each image patch to match the audio, using a loss:

$$\begin{aligned}\mathcal{L}_{IS}(\tilde{A}_{IS}) &= \mathcal{L}_{CE}(\tilde{A}_{IS}, \mathbf{y}_{IS}) \\ &= -\frac{1}{N}\sum_i \log \tilde{A}_{IS}(i, \mathbf{y}_{IS}(i)),\end{aligned} \tag{2}$$

where $\mathcal{L}_{CE}$ is cross-entropy loss and $\mathbf{y}_{IS}$ is a label vector that indicates which sound each image patch corresponds to. This model is similar to [3], and can be interpreted as a random walk that starts at an image patch and ends at an audio example, and which is judged to be successful if it ends at the audio that corresponds to the image patch. One can also learn a similar random walk that starts at the audio, and obtains credit if it ends at any patch from its corresponding image:

$$\mathcal{L}_{SI}(\tilde{A}_{SI}) = -\frac{1}{N}\sum_i \log \sum_j \tilde{A}_{SI}(i, \mathbf{y}_{SI}(i,j)), \tag{3}$$

where $\tilde{A}_{SI}$ is similar to $\tilde{A}_{IS}^\top$ but with softmax normalization on its rows, and $\mathbf{y}_{SI}(i,j)$ is used to index all the patches corresponding to i-th sound.

Visual similarity. Following other self-supervised approaches, after training with either Eq. (2) or Eq. (3), we can then measure the similarity between patches by comparing the learned representation: two patches "go together" if they make the same sound. This similarity score between two patches can be computed as $\tilde{A}_{IS}\tilde{A}_{IS}^\top$, by which the visual similarity is measured by audio-visual similarity. Under this model, the similarity between two image patches is given as the probability of a cross-modal random walk beginning at one patch and ending at another.

Sound source localization. By solving the audio-visual correspondence task, the model can learn to localize sound sources [4, 33, 28]. This is achieved by computing a confidence map using the audio and visual features corresponding to different spatial locations, *i.e.*, the confidence map that localizes the sound source k can be computed by $\tilde{A}_{SI}(k)$.

However, this approach for learning a similarity function has limitations. First, the grouping is defined entirely based on local information inside each patch. Second, there is no explicit constraint that patches on the same object be grouped together: as pointed out by [25] the loss only encourages cross-modal similarity, and relies on it to induce within-modality similarity indirectly. Consequently, if features are learned entirely by cross-modal correspondence, many patches on the objects won't be discriminative enough for telling which audio to match, *e.g.*, telling whether a person's clothing should be grouped with their faces.

3.2. Learning grouping with random walks

We propose to address these shortcomings by extending the random walk formulation in two ways: 1) by adding cycle-consistency constraints that explicitly creates within-modality similarity, and 2) by incorporating non-local information into the similarity metric to allow for matching less-discriminative regions. The complete model is shown in Figure 2.

Random walk with cycle-consistency. We'll measure the similarity between patches as a path-finding problem and use cycle-consistency loss to add explicit constraints that encourage learning within-modality similarity. The structure of this graph resembles that the one in Section 3.1. We construct a directed graph, where nodes are patches or audio, and weights of edges connecting nodes are dot products of visual and audio features. A random walker in this graph that starts from a patch node $\mathbf{x}_i$, travels to an audio node $\mathbf{s}_k$ and returns to another patch $\mathbf{x}_j$. As before, we measure the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$ as the probability of walking from $\mathbf{x}_i$ to $\mathbf{x}_j$, *i.e.*,

$$\tilde{A}_{ISI}(i,j) = (\tilde{A}_{IS}\tilde{A}_{SI})(i,j). \tag{4}$$

We'd like to add an explicit constraint that encourages $\tilde{A}_{ISI}(i,j)$ to be large if $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same object. This seemingly leads to a chicken-and-egg problem: how can we identify whether a pair of patches are on the same object, if we do not know what the objects are? Inspired by recent works on visual correspondence [39, 21], we use a *cycle-consistency* constraint. While we do not know whether $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same object, we *do* know—trivially—that $\mathbf{x}_i$ and $\mathbf{x}_i$ are on the same object. Therefore, we encourage patches to return to themselves with high probability. Specifically we optimize:

$$\mathcal{L}_{cyc}(\tilde{A}_{ISI}) = \mathcal{L}_{CE}(\tilde{A}_{ISI}, \tilde{I}), \tag{5}$$

where $\tilde{I}$ is the column index of diagonal elements in an identity matrix. This indicates an image patch should be able to successfully return to itself on a random walk after walking to the audio. Since the random walk is memory-less, the walk that originates from an audio node will lead to the same set of visual nodes. Due to the cycle-consistency loss, this walk is encouraged to maximize the probability of reaching all of the visual nodes that can reach the audio. This is in contrast to a model that optimizes $\mathcal{L}_{SI}$ alone, which would allow for the walk to reach *any* image patch. In this way, it encourages patches on the same object to go together and hence benefits grouping.

Within-modality transitions. The loss in Eq.(5) still only learns audio-visual correspondences. This limits the capability of the model to learn less discriminative regions, which are less likely to walk to the correct audio. For example, connecting a patch on the human body with the voice of the speaker is challenging. We address this by modeling transitions *between* image patches as well. This allows the model to learn similarity through an indirect transition: *e.g.* walking from the body to a face, then from the face to the audio. We factorize $\tilde{A}_{IS}$ as $A_{II}A_{IS}$, and $\tilde{A}_{SI}$ as $A_{SI}A_{II}$, where A_{II} denotes the affinity matrix of walking from a patch to another, *i.e.*, the transition from $\mathbf{x}_i$ to $\mathbf{x}_j$ is

$$A_{II}(i,j) = \frac{\exp(f(\mathbf{x}_i)^\top f(\mathbf{x}_j)/\tau)}{\sum_k \exp(f(\mathbf{x}_i)^\top f(\mathbf{x}_k)/\tau)}. \tag{6}$$

In Fig. 2, we show several possible paths for a patch to walk back to itself in the graph. Combining this idea with cycle-consistency, the loss can be reformulated as

$$\mathcal{L}_{cyc}(A_{cyc}) = \mathcal{L}_{CE}(A_{cyc}, \tilde{I}), \tag{7}$$

where the cycle matrix A_{cyc} is

$$A_{cyc} = A_{II}A_{IS}A_{SI}A_{II}. \tag{8}$$

Combining these ideas together, our random walk model is trained using both correspondence losses, which exploit known audio-visual correspondences, and cycle loss, which encourages within-modality similarity:

$$\mathcal{L} = \mathcal{L}_{cyc}(A_{cyc}) + \mathcal{L}_{IS}(A_{IS}) + \mathcal{L}_{SI}(A_{SI}). \tag{9}$$

Transitions to other examples. In addition to allowing image patches to transit to other (non-corresponding) audio tracks, we also allow patches to jump to other images, to encourage learning robust visual features. The resulting patch similarity, described by A_{II}, should provide paths connecting different parts of the object. For example, in the second half audio-to-image cycle path, A_{II} routes the random walk from the part of the image that distinctively indicates the sound (*e.g.* a mouth) to the rest of the object. Note that many image patches will not be able to obtain cycle consistency when allowing transitions to other examples. For example, there is little information for a patch sampled from the background of a scene to guide itself to the audio it corresponds to. As a result, these patches can only obtain low probability for cycle paths, *i.e.*, the magnitude of $A_{cyc}(i,i)$ could be very low.

Localizing sound sources. We can take advantage of the within-modality similarity metric to learn sound source localization models that segment a larger portion of the object than previous approaches. Similar to other self-supervised approaches, we localize the sound source using ground-truth correspondence: namely, the paired image and audio examples. For the k-th audio in the batch, $A_{SI}(k)A_{II} \in \mathbb{R}^N$ can be used to get the sound source segmentation, *i.e.*, the probability of patches being part of the object.

3.3. Selecting cycle paths

Limiting transitions between images. Since there are multiple patches per image, but only one audio, there will be multiple paths that converge on the same audio node. Consequently, the walk exiting an audio node will be responsible for routing the walk to a large number of nodes, each of which has a small transition probability. To improve optimization, for the second half audio-to-image cycle, we limit

the transition and only allow walking to paired audio and images. We modify the affinity matrix A'_{SI} such that the probability of paths between unpaired audio/image pairs have been zeroed out and re-normalized. We then use an affinity matrix A'_{II} that only allows walks to patches that belong to the corresponding image. We call this the *dead end* formulation. The new cycle matrix will be $A'_{cyc} = A_{II}A_{IS}A'_{SI}A'_{II}$.

Ignoring background patches. The cycle loss in Eq.(7) encourages every patch to complete a cycle. However, this can be problematic, because many patches in the background of the scene will be unable to reach the audio, yet are encouraged to complete a cycle. This can lead to a model that enforces many background patches to assign a relatively large amount of probability mass to all patches in an image, whether they are on the foreground or background. We therefore propose to apply the cycle loss only to patches that have high probability of reaching the correct audio. We use the probability matrix A_{SI} to select only the high-probability patches. We do this by using only the $\beta \cdot N$ patches with highest probability of reaching the audio, where $0 < \beta < 1$ is a hyper-parameter.

4. Experiments

4.1. Implementation

Audio-visual graph. We sample 7×7 patches of size 64×64 from 256×256 images, following [21], and 0.96s of audio. Each image-audio pair contains 50 nodes, including 49 patch nodes and 1 audio node, and we use 16 pairs to construct an audio-visual graph. We add random cropping to every patch to avoid the network utilizing border information, following [21]. We randomly shift audio for 0 – 0.3s for augmentation. At the test time, we use 32×32 patches to generate a denser graph and get results with higher resolution.

Patch encoder. We use ResNet18 [17] as the backbone for the patch encoder. Each patch will be encoded as a 128-dimensional feature vector with l_2 normalization following [40]. We use 2 different embeddings for computing A_{II}, *i.e.*, the weight of the edge connecting $\mathbf{x}_i$ and $\mathbf{x}_j$ is computed using $f_1(x_i)$ and $f_2(x_j)$.We also learn another encoder f_3 for measuring audio-visual similarity. We compute all embedding vectors using different fully connected layers on the final layer of the network backbone.

Audio encoder. We adopt VGGish network[18] to extract l_2 normalized 128-dimensional audio features from log-mel spectrograms, which are converted from 0.96s raw waveform by applying short-time Fourier transform.

Parameter. During training, we use the Adam optimizer [22] with a learning rate $= 10^{-4}$, with a batch size of 64 and temperature $\tau = 0.07$ following [40]. β is set to 0.3 for both datasets. Please see the supplementary for more training details.

4.2. Datasets

Human speech. VoxCeleb2 dataset [8] is a large-scale audio-visual speaker recognition dataset containing over 1.5k hours of videos for 6,112 celebrities. We generate segmentation masks by applying Mask R-CNN [16] for 1k random samples in the test set.

Musical instruments. We use VGGSound dataset [5] which contains over 550 hours of video clips for 311 different classes. For evaluation, we filtered and annotated segmentation masks for 446 high-quality video frames. There is an ambiguity in defining what the object is since usually people and instruments co-occur in the scene, and there is not a clear delineation between them. Therefore, when evaluating VGGSound-Instruments, we also create a second evaluation set that includes human body masks as part of the ground-truth object.

4.3. Evaluated methods

We compare our model with several other methods. For each model, we reproduce the results using ResNet18 with both feature map size (before fully connected layers) of 16×16 (similar to original method) and 32×32 (similar to our method) as the backbone to ensure a fair comparison.

Self-supervised audio-visual methods. We compare with several variations of the model from [4]. To decouple the contribution of our proposed loss from the contrastive learning technique, we use an InfoNCE [26] for the contrastive loss, following our own model. We follow [4] to implement AVOL-GMP, which uses a global max pooling layer. And we also implement AVOL-GAP with global average pooling layer, which is resemble to the strategy used in [2]. In contrast to approaches that use multiple frames [1, 2, 31, 45] or require extra manual labels [19, 30], these methods only use one frame and are trained in a fully self-supervised way.

Weakly supervised localization method. We also consider the class activation map (CAM) [46] model, which uses image-level class labels in a weakly supervised way. Recently, this method has been shown to obtain comparable results to most state-of-the-art methods in weakly supervised object localization (WSOL) [6]. We only train CAM on the instruments dataset, because there are no explicit category labels for the VoxCeleb2 speech datasets.

Grouping methods. COOC [20] is a self-supervised visual grouping method that uses the information from co-occurrence in space. It learns whether a pair of image patches are (or are not) spatially adjacent, and has been applied to segmentation tasks. For an apples-to-apples comparison, we use the same patch size and stride as our method, and use an InfoNCE contrastive loss. We also train a variation of this method, COOC-Img, by sampling positive patch pairs from same images and negative pairs from different images.

Supervised features. We compare with features of a CNN

Dataset	VoxCeleb2	VGGSound-Instruments	
Object	Ppl.	Instr.	Ppl. & Instr.
ImageNet* [17, 9]	72.2	75.7	71.5
CAM(16×16)* [46]	–	**87.0**	**82.7**
CAM(32×32)* [46]	–	77.0	71.9
COOC [20]	57.1	77.8	72.6
COOC-Img	77.4	79.8	78.8
AVOL-GAP(16×16) [2]	66.7	77.2	75.3
AVOL-GAP(32×32) [2]	48.0	76.6	71.0
AVOL-GMP(16×16) [4]	48.0	53.8	54.1
AVOL-GMP(32×32) [4]	43.7	51.5	53.7
IS	71.6	72.2	74.1
SI	72.5	79.6	68.6
IS+SI	77.0	81.6	71.9
Ours	**83.8**	**87.0**	**79.7**

Table 1: Evaluation of same-object grouping. We evaluate the average precision score at predicting whether two random patches are on the same object. $*$ indicates methods trained with image-level labels.

trained with object labels on ImageNet [9]. We use the network to extract features for each image patch and then take the l_2 normalization following our implementation.

Center prior. We also use a simple *center* baseline, which ignores the image and predicts a score-map with a peak at the image center. Specifically, our scores are given as the probability density function of a Gaussian centered on the image [6] with a standard deviation equal to 100.

4.4. Evaluation of grouping

We evaluate these grouping methods on two criteria: (a) *same-object grouping*: are methods capable of distinguishing a pair of patches that are both on the same object, from a "mismatched" pair on different objects? (b) *completeness*: are methods capable of grouping as many parts of the object as possible? A good grouping method should perform well under both metrics.

4.4.1. Same-object grouping

Evaluation metric. We first evaluate the same-object grouping following [20]. Given two patches sampled on/off the object, we compute the average precision (AP) of whether the model can distinguish these two categories. We randomly sample pairs of patches on testing images. Positive pairs are constructed by two patches on the object (at least 50% of the pixels are on the object), while the negative pairs contain one patch on the object and the other on the background.

Baselines setup. To demonstrate cycle-consistency does encourage better visual grouping, we compare with several variations of our method, as discussed in Section 3. 1) **IS** model, which is trained with Eq. (2). 2) **SI** model, which is trained with Eq. (3). 3) **IS+SI** model, which is trained with both correspondence losses. These three models above only learn similarity from the correspondence of image-audio pairs.

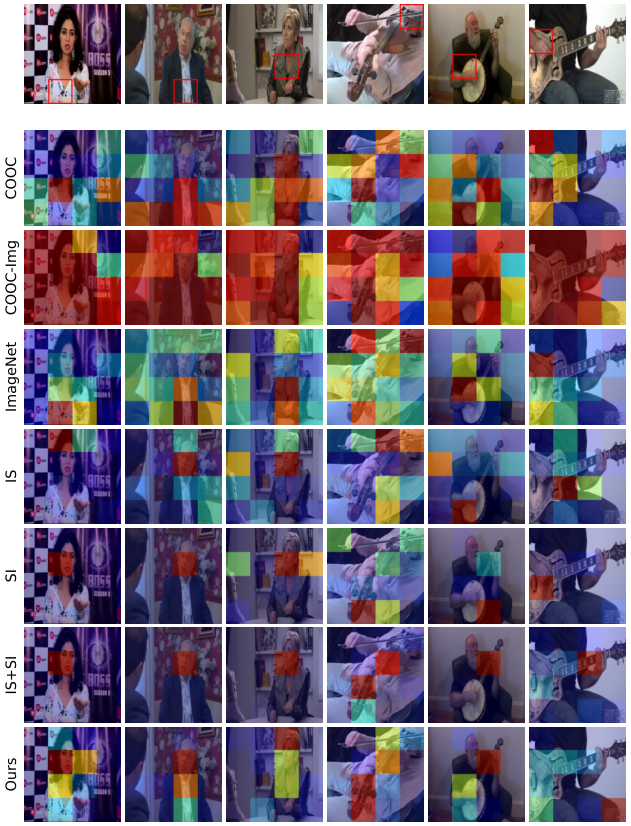


Figure 3: **Same-object grouping.** For a given input patch (top), we measure similarities to other image patches in a 4×4 grid. The color of the patch indicates its similarity (red means high, blue means low). Please see the supplementary for more results.

Quantitative results. For IS, SI, and IS+SI models, we use Eq. (4) to predict similarity scores. For our full model, we directly use the cycle matrix A_{cyc}, *i.e.*, the probability of random walk from one patch to another, as the similarity score. For other methods, we compute the similarity score using the dot product between visual features. The results are shown in Table 1. It can be seen that our full model outperforms other methods trained only with audio-visual correspondence, including variants of our model and other audio-visual approaches. This suggests that cycle-consistency is a useful constraint for learning similarities between different parts of an object. We notice that our method learns a feature set that achieves a competitive performance with pre-trained model on ImageNet as well as CAM, which are trained in a supervised way.

Qualitative results. In Fig. 3, for each image, we divide it into 4×4 grids. We randomly select one patch on the object, then visualize the similarity scores between this patch and other patches. We can see that methods only learning cross-modality similarity fail to learn the similarity of different parts from one object, which indicates lower grouping capac-

Dataset		VoxCeleb2			VGGSound-Instruments					
Object		People			Instruments			People & Instruments		
		PxAP	pIOU	mIOU	PxAP	pIOU	mIOU	PxAP	pIOU	mIOU
	RAG+NCut [38, 34]	45.8	42.8	42.8	25.2	24.2	24.2	50.3	46.5	46.5
	ImageNet* [17, 9]	**65.6**	**57.4**	46.6	**40.0**	**36.7**	24.7	63.4	55.5	47.8
	COOC [20]	59.0	49.8	45.8	32.5	33.4	26.3	58.1	54.3	50.8
	COOC-Img	63.8	56.7	**49.5**	36.7	35.0	**26.8**	**64.2**	**56.7**	**51.1**
	Center prior [6]	62.2	47.3	43.6	38.5	30.3	25.2	69.6	53.6	**48.1**
WSOL	CAM*(16×16) [46]	–	–	–	50.5	39.1	30.1	70.3	55.1	47.0
	CAM*(32×32) [46]	–	–	–	51.8	40.4	33.1	67.0	52.6	44.1
Self-supervised audio-visual learning	AVOL-GAP(16×16) [2]	47.4	44.7	30.0	31.7	29.8	5.8	55.0	50.4	22.5
	AVOL-GAP(32×32) [2]	48.4	43.7	31.5	35.5	32.2	5.3	59.4	51.4	20.6
	AVOL-GMP(16×16) [4]	53.6	46.3	37.2	50.9	38.6	15.9	71.6	54.9	31.7
	AVOL-GMP(32×32) [4]	56.8	47.4	32.0	**57.5**	42.4	23.2	**72.1**	54.9	25.3
Ours	IS	51.7	46.6	37.8	35.6	32.8	24.6	59.0	53.2	45.8
	SI	63.4	49.5	41.3	50.5	41.3	32.7	68.0	55.5	47.3
	IS+SI	62.4	49.1	39.8	50.1	41.4	33.4	67.5	55.6	47.1
	Ours	**65.8**	**50.7**	**43.7**	56.2	**44.5**	**34.1**	70.8	**56.7**	47.2

Table 2: Evaluation of grouping completeness. We evaluate with PxAP, pIOU, and mIOU to ensure threshold independence. $*$ indicates methods trained with image-level labels. The first four methods take advantage of normalized cuts, because we take the cut with higher score as foreground since they cannot identify which cut belongs to objects.

ity. For example, when sampling from less discriminative parts of the object, like the arm of the person, it usually only has a higher similarity with the most discriminative parts, like the mouth. In contrast, our method obtains closer to uniform similarity with all patches on the object, indicating better grouping performance.

4.4.2. Completeness of grouping sound source

Evaluation metric. We evaluate completeness of grouping a sound source using the metrics for object segmentation following weakly supervised localization works [43, 6]. The evaluation metrics for weakly supervised localization are disputed [6]; it is believed that current state-of-the-arts take advantage of the fixed threshold for score-maps. Taking it into consideration, we use three metrics to evaluate completeness of grouping. Inspired by [6], we use pixel-wise average precision (**PxAP**) and averaged best IoU per image (**pIOU**), which selects the best threshold for every sample, for threshold independence. We also use a conventional metric, mean Intersection over Union (**mIoU**), which selects the threshold for score-maps by observing 50 samples.

Quantitative results. Using audio-visual correspondence enables us to obtain segmentation for sound source easily, *i.e.*, we localize object using A_{SI} or $A_{SI}A_{II}$. For those methods based on affinity matrix between patches, we use normalized cuts [34] to get 2 segmentations, which can be viewed as binary score-maps, and the one with higher score is identified to be foreground. Then we can upsample score-maps to the same size as the original image. We use the same upsampling strategy for all patch-based methods to ensure fair comparisons; the score of one pixel on the upsampled score-maps is computed by taking the average score of patches covering that pixel. The results are shown in Table 2. Comparison with variations of our model indicates the proposed cycle-consistency and within-modality transitions benefit grouping. Our method achieves better performance on VoxCeleb2, probably because objects like people usually contain less discriminative regions for finding an audio. Our performance is close to AVOL-GMP, COOC-Img on several metrics. We note that a method doesn't need to perform well on same-object grouping while achieving good scores on completeness, *e.g.*, the operating threshold could be low enough to classify less discriminative parts as an object, but these parts could have higher similarities with background on the image. Hence, in general, by jointly observing Table 1 and Table 2, our method outperforms other methods.

Qualitative results. In Fig. 4, we visualize the score-maps generated by these methods. It can be seen that models trained with only audio-visual correspondence and CAM mainly focus on the most discriminative parts of the object, while our method spreads the probability to the whole object.

Conclusively, models based only on the audio-visual correspondence cannot perform well on grouping considering same-object grouping and completeness metrics, and the proposed cycle-consistency and within-modality transitions can improve grouping performance significantly.

4.5. Ablation study

Ignoring background patches. We design ablation experiments for hyper-parameter β by setting it to ten different values. From Fig 5, we can see that when $\beta \leq 0.5$, it does not have much impact on our performance, while the performance gradually drops as β increases from 0.5 to 1.0. This

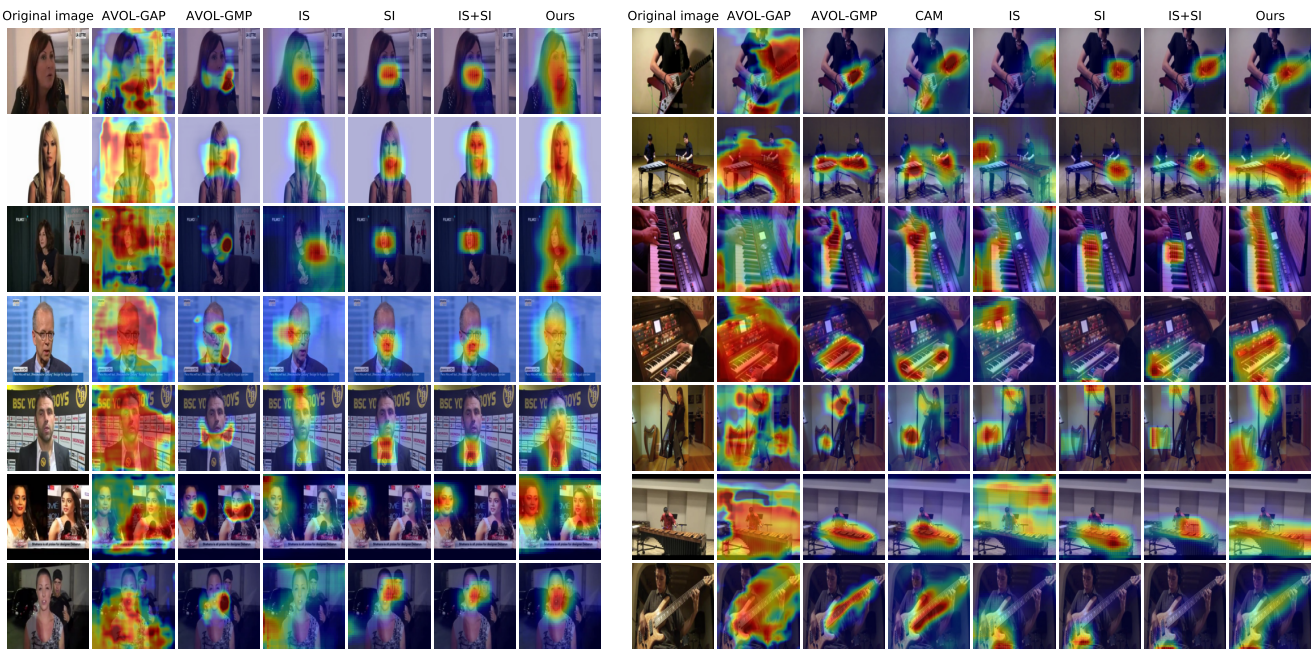


Figure 4: **Completeness of grouping sound source.** Comparison of score-maps generated by different methods. Red for higher score and blue for lower. The results in the last three rows are randomly sampled. Please see the supplementary for more results.

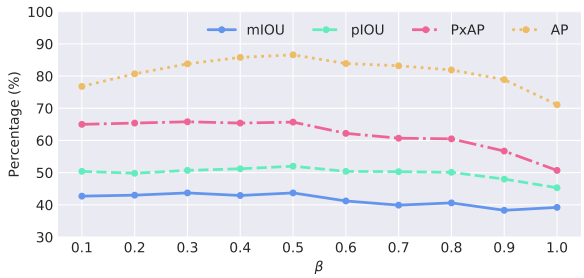


Figure 5: The effect of hyper-parameter β. Evaluations are conducted on the VoxCeleb2 with four grouping metrics, *i.e.*, PxAP, pIOU, mIOU, and AP on same-object grouping.

Dead end	I-I trans.	AP	PxAP	pIOU	mIOU
✓		84.5	57.4	49.4	41.2
	✓	84.0	63.7	49.8	40.2
✓	✓	**85.8**	**65.4**	**51.2**	**42.9**

Table 3: Ablation results for dead end and within-modality transitions on VoxCeleb2 dataset. AP denotes average precision metric for same-object grouping. *I-I trans.* denotes within-modality transitions.

is because the proportion of negative background patches are increasing, which can be harmful to our grouping results. This result confirms that it is reasonable and helpful to use β to ignore background patches in our method.

Dead end. We compare our method with models that no dead end is applied. We train both models on the VoxCeleb2 dataset with $\beta = 0.4$. As can be seen from Table 3, the model with dead end outperforms the one without, suggesting that simplifying the cycle paths could benefit our method.

Within-modality transitions. We compare with a model described in Section 3.3 that is trained without allowing within-modality transitions, and it uses the same method for selecting cycle paths ($\beta = 0.4$) as our full model. In Table 3, it can be seen that our full model performs better than this model, indicating that within-modality transitions help learning features for less discriminative regions. Interestingly, this model outperforms other methods trained only with audio-visual correspondence on same-object grouping, confirming cycle-consistency helps visual grouping.

5. Conclusion

We propose a method achieving better visual grouping performance using random walk with cycle-consistency constraints. Our method is able to learn better within-modality similarities, leading to improved visual grouping. We show that our method identifies and segments sound sources better than methods based on audio-visual correspondence learning alone. We hope this paper provides further directions for the study of multi-modality learning. The first is to develop new methods for utilizing multi-modality signals for grouping tasks. The second direction is to combine methods for weakly supervised with class labels with the rich signal provided by audio. We will release code, pre-trained models, and our datasets.

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